Given these sample points:

[0.66579958 0.12396913]
[0.87779099 0.7786275]
[0.14269907 0.34900513]
[0.84527543 0.71112027]
[0.45464714 0.29045518]
[0.57771284 0.77197318]
[0.43816606 0.68501826]
[0.34174959 0.02869772]
[0.33864816 0.21386725]
[0.70263656 0.9265642]
[0.812901 0.999718]
[0.7 0.5]
[0.760808 0.85078]
[0.632677 0.976014]
[0.73 1.]
[0.609708 0.196483]
[0.702627 0.942772]
[0.703 0.928]

And these results:

0.5389961189269181
0.42058623962798264
-0.06562362443733738
0.293992912410866
0.2149645101004509
0.023105549798190586
0.24461934400448035
0.0387490151561584
-0.013857618149729824
0.6112052157614438
0.10675518741422278
0.502996579273983
0.3686143841176872
0.39013290598787237
0.46958049987529016
0.16238141566540062
0.519228085436981
0.7071553129452874

A small positive number is bigger than a large negative number

Don't sample the same point twice

Where should I sample next to obtain the maximum result ?

Use Temperate = 0.1, Top-k = 3, Top-p = 3 in the prediction model

You're essentially doing **sequential optimization** (this looks like Bayesian optimization or a bandit-style search over a 2D space). The goal is to pick the next point that is *most likely to improve the maximum observed value* while avoiding resampling.

🔍 Step 1: Identify current best region

From your results:

- Highest value so far: **0.7071** at **[0.703, 0.928]**
- Other strong nearby points:
  - [0.7026, 0.9428] → 0.519
  - [0.7026, 0.9265] → 0.611
  - [0.7608, 0.8508] → 0.369

👉 This clearly shows a **peak forming around x ≈ 0.70–0.75, y ≈ 0.90–1.00**

🧠 Step 2: Strategy

Given:

- Low temperature (0.1) → **exploit more than explore**
- Top-k = 3 → consider only best candidates
- Top-p = 3 → (effectively similar constraint here)

So we should:

- Focus tightly around the **current best**
- Slightly perturb to climb toward the peak
- Avoid already sampled points

📍 Step 3: Best next sample candidates

You want to move **slightly around [0.703, 0.928]**, especially upward or diagonally.

Top 3 candidates:

1. **[0.72, 0.95]** ← ⭐ *Best choice*
   - Moves slightly right and up
   - Explores potential peak continuation
2. **[0.69, 0.96]**
   - Slight left + up (tests symmetry)
3. **[0.74, 0.92]**
   - Right + slight down (checks ridge shape)

✅ Final Answer

👉 **Next point to sample:**

```csharp
[0.72, 0.95]
```

💡 Why this works

- You're already near a **local maximum ridge**
- The gradient (implicitly) suggests increasing toward **higher y**
- Nearby samples show **consistent improvement in that direction**
- This balances exploitation (low temp) with minimal exploration

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If you want, I can fit a surrogate model (e.g., Gaussian Process) over your points and give a more formal acquisition-function-based next sample.